less common than that. And yet, they only match 25 percent of the time under those circumstances. The inspectors are forced to conclude that the roulette balls are not sharing instruction sets. Yet the roulette balls always land on matching colors when Gillian and Fatima are using corresponding roulette wheels, so they clearly do have some kind of coordination going on—that's what led the inspectors to suspect the balls were sharing instruction sets to begin with. Therefore, to account for these results, the roulette balls must be sending signals to each other after they know which wheel they're arriving at.

The preceding section is a proof of Bell's theorem, thinly disguised. The pairs of roulette balls are pairs of photons with entangled polarizations. The roulette wheels are polarizers that measure polarization along three different directions, selected randomly while the photons are in flight toward the polarizers. And Bell's theorem is the proof embedded in the story, the one that Fatima figured out. If your roulette balls really behave that way, there's something strange going on, and it can't be accounted for by assuming the roulette balls have hidden instructions—hidden variables—that they carry with them from the moment they separate. And entangled photons really do behave this way, so something very strange must be going on in quantum physics. But what, exactly, did Bell prove? To understand this, let's take a closer look at what happened at Ronnie's casino.

We started out with the assumption that roulette balls can't magically communicate with each other instantaneously across long distances (though we never explicitly stated this until the end). In other words, we started out with the assumption of locality. This led us to the idea that there must be hidden instruction sets in the roulette balls themselves, because that was the only way to account for the perfectly matched outcomes when Gillian and Fatima used the same roulette wheels. But the strange correlations in outcomes when Gillian didn't use the same wheel as Fatima ruled out the possibility of hidden variables. Therefore, something must be wrong with our assumption: locality must be violated. In the case of Ronnie's casino, the roulette balls could still be communicating by radio, of course. But, in